

FULL MARKS: 80
TIME: 2 HOURS

*Attempt all questions from Section I and any four questions from Section II.
The intended marks for questions or parts of questions are given in brackets [].
This question paper consists of 6 printed pages.*

Question 1.

- i. A cell has five pairs of chromosomes. After mitotic division, the number of chromosomes in the daughter cells will be:**

 - a. Five,**
 - b. Ten,**
 - c. Twenty,**
 - d. Forty.**
- ii. The specific function of light energy in the process of photosynthesis is to:**

 - a. Reduce Carbon dioxide,**
 - b. Synthesise glucose,**
 - c. Activate chlorophyll,**
 - d. Oxidise Carbon dioxide.**
- iii. The rate of photosynthesis is not affected by:**

 - a. Light intensity,**
 - b. Humidity,**
 - c. Temperature,**
 - d. Carbon dioxide.**
- iv. In cohesion-tension theory, cohesion refers to the:**

 - a. Attraction between water molecules**
 - b. Attraction between the xylem vessel**
 - c. Attraction between water molecules and xylem**
 - d. Attraction between water molecules and phloem.**
- v. Insulin is secreted by**

 - a. Beta cells of Pancreas,**
 - b. Alpha cells of Pancreas,**
 - c. Delta cells of Pancreas,**
 - d. Gamma cells of Pancreas**

- vi. The crossing of a pure Tall, Round seed plant and a pure Dwarf, Wrinkled seed plant was done. What will be the phenotypic ratio of F₂ generation?
- 1:2:1,
 - 9:3:3:1,
 - 3:1,
 - 9:2:3:1
- vii. If a pure tall plant is crossed with a pure dwarf plant, the phenotype of the off springs of F₁ generation will be:
- All Tall,
 - All Dwarf
 - 3 Tall and 1 dwarf,
 - 50% Tall and 50% Dwarf
- viii. The chromatids of a chromosome are attached at a point called:
- Centriole,
 - Centromere,
 - Chromomere,
 - Centrosome
- ix. The main impact of greenhouse effect is:
- To form acid,
 - Global warming
 - Noise pollution,
 - Thermal pollution
- x. Which of the following hormone is used for ripening of banana: commercially?
- Ethylene,
 - Cytokinin
 - Absciscic acid,
 - Gibberellins
- xi. Compound formed when carbon dioxide combines with haemoglobin:
- Carbaminohaemoglobin
 - Carboxyhaemoglobin
 - Oxyhaemoglobin
 - Carbohaemoglobin
- xii. Contraction of uterus during child birth is due to:
- TSH
 - Glucagon
 - Oxytocin
 - FSH
- xiii. Which one of the following refers to a normal human female?
- 22 and XX
 - 22 and XO
 - 23 and XX
 - 22 and XY
- xiv. Body fluid occupying spaces between the cells in the organ:
- Tissue fluid
 - Blood
 - Lymph
 - Organ fluid
- xv. Duplication of chromosome occurs during:
- Interphase
 - Prophase
 - Mitosis
 - Cytokinesis

Question 2

(i) Name the following:

[5]

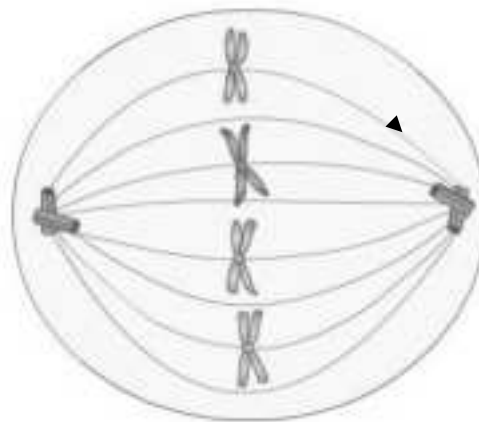
- The set of genes present in the cells of an organism.
- A blood group which is known as a universal donor.
- The colour of the blood when it is taken from the artery.
- A gaseous plant hormone.

- e. The part of the chloroplast where the light reaction of photosynthesis takes place.
- (ii) Choose the odd one out and write the correct category to which the others belong: [5]
 Example- Large Intestine, Stomach, Mouth, Brain.
 Odd one- Brain, Category- Organs of Digestive System
- Thyroid, Thyroxine, Pancreas, Adrenal Gland
 - Oxytocin, Ethylene, Cytokinin, Auxin
 - Goitre, Cretinism, Myxoedema, Addison's Disease
 - Needles, Syringe, Dirty dressings, Cobalt 60
 - Glycolysis, Photosynthesis, Photolysis, Phosphorylation
- (iii) Arrange and rewrite the terms in each group in the correct order so as to be in a logical sequence beginning with the term that is underlined. [5]
- Soil water, Xylem, Root hair, Cortex.
 - Refrigerator, Skin cancer, Ozone layer depletion, CFCs.
 - Living cell, Flaccid, Hypertonic, Exosmosis.
 - Transpiration, Ascent of sap, Suction force, Root cell.
 - G₁ phase, Mitosis, S phase, G₂ phase.
- (iv) Match the items given in Column I with the most appropriate ones in Column II and rewrite the correct matching pairs. [5]
- | Column I | Column II |
|------------------------|------------------------------|
| a. Fibrinogen | a. Smog |
| b. Respiratory pigment | b. Water Pollution |
| c. Osmosis | c. Urine |
| d. Urochrome | d. Haemoglobin |
| e. Muslin cloth | e. Blood |
| | f. Semi permeable membrane |
| | g. Freely permeable membrane |
- (v) Mention the exact location of the following: [5]
- Thylakoid
 - Ureter
 - Pituitary Gland
 - Bowman's Capsule
 - Thyroid Gland

SECTION B (40 Marks)
(Attempt any four questions from this section)

Question 3

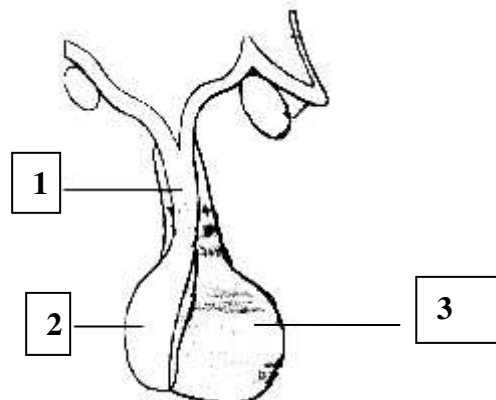
- (i) Define Tropic hormone. [1]
- (ii) Mention the name of the compound formed when carbon monoxide combines with haemoglobin. Why is carbon monoxide harmful for your body? [2]
- (iii) A Tongue roller mother (Tt) and a Non tongue roller father (tt) produces offsprings. Find out the percentage of offsprings who are tongue roller and percentage of offsprings who are non tongue roller. [2]
- (iv) Explain the term Mutation. Name a disease caused in human beings due to gene mutation. [2]
- (v) Observe the diagram given below and answer the following questions: [3]



- a. Identify the stage shown in the above diagram and give a reason to support your answer.
- b. Mention one important feature of the next phase.

Question 4

- (i) Define karyotype. [1]
- (ii) Observe the diagram below and answer the following questions: [2]

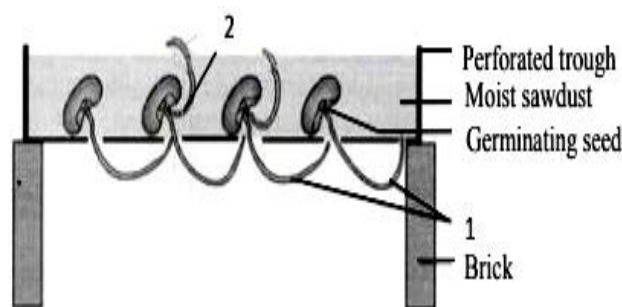


- a. Identify the Endocrine gland shown in the above diagram.
- b. Mention the full form of ACTH.
- (iii) Mention two precautions in the use of a potometer. [2]

- (iv) “Mature Red Blood cells lack nucleus”. Give two reasons to justify this statement. [2]
- (v) Mention one important function of the following: [3]
- Chloroplast
 - Cytokinin
 - Glucagon

Question 5

- (i) What do you understand by the term Leukopenia? [1]
- (ii) Why do we keep a leaf in boiling water during starch test? What do you understand by the term destarching? [2]
- (iii) Mention any two reasons to state why Mendel had selected Garden Pea for his experiments. [2]
- (iv) Define Parthenocarpy and give one example of a parthenocarpic fruit. [2]
- (v) Study the diagram given below and answer the following questions: [3]



- Identify the tropic movements shown by “1” in the diagram.
- Which tropism is effective here in case of ‘1’?
- Identify the tropic movement observed for ‘2’.

Question 6

- (i) Define pollution. [1]
- (ii) Mention two objectives of Swachh Bharat Abhiyan. [2]
- (iii) Mention two harmful effects of acid rain on monuments and buildings. [2]
- (iv) Differentiate between Mitosis and Meiosis on the basis of [2]
- Number of daughter cells produced.
 - Site of occurrence
- (v) Pure Green pod (GG) pea plant is crossed with Pure Yellow pod (gg) pea plant. [3]
- Mention the phenotypic and genotypic ratio of F₂ generation.
 - Define Mendel’s Law of Dominance.

Question 7

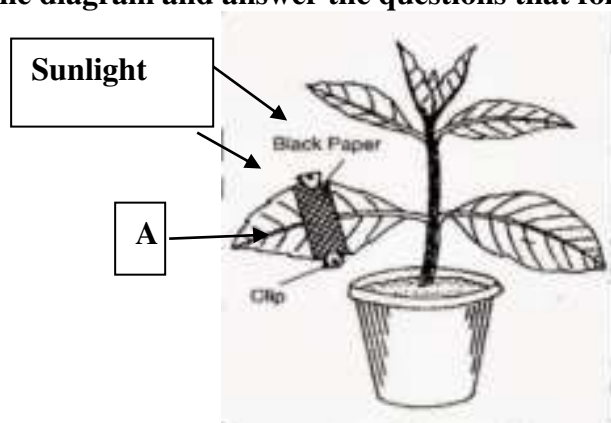
- (i) Mention the scientific name of the plant with which Mendel had conducted his experiments. [1]
- (ii) State briefly about the following abnormal conditions in human body. [2]
 - a. Haematuria
 - b. Glycosuria
- (iii) What do you understand by the term “Destarched Plant”? How is a plant destarched? [2]
- (iv) Define Photosynthesis and write the balanced equation of the process. [2]
- (v) Answer the questions that follow: [3]
 - a. Why are platelets known as thrombocytes?
 - b. Draw a flowchart to show the process of blood clotting when a person is injured.

Question 8

- (i) Mention the full form of ABA. [1]
- (ii) Answer the following questions: [2]



- a. Identify the pollution shown in the diagram given above.
- b. List any two sources of this pollution.
- (iii) Explain the term Chemotropism and give one example of this tropic movement. [2]
- (iv) Study the diagram and answer the questions that follow: [2]



- a. State the aim of the experiment.
- b. Mention what do you observe at the end of the experiment when the leaf labelled 'A' is tested with iodine solution.
- (v) Draw a neat labelled diagram of a chloroplast. [3]